# HELIOS / EV_OS: Comprehensive DIY Hypercar Specification & Architecture Master Blueprint

This document serves as the master engineering reference for the **HELIOS** project repository (EV_OS). It consolidates the vehicle concept, software architecture, complete hardware part lists, electrical/thermal topologies, and safety requirements into a single integration specification designed for workspace ingestion via VS Code Copilot.

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## 1. Executive Summary & Vehicle Concept

HELIOS is an open-architecture, software-defined electric and hybrid hypercar concept utilizing a **Zonal Electrical Architecture** and a **Triple-Hybrid High-Voltage (HV) Topology**.

### Performance Targets (Unvalidated Simulation Goals)

- **Range Target:** > 500 miles
- **0-60 mph Target:** < 2.0 seconds
- **Peak Power Target:** > 1500 kW
- **Active Downforce Target:** Up to 2000 kg via Active Ground-Effect Downforce System (AGEDS)

### Powertrain Architecture & Subsystems

1. **Hydrogen Fuel Cell (HFC):** 10 kW M240 metallic stack (Hyfindr) backed by a 700-bar 218L Type 4 cylinder, Kawasaki Heavy Industries high-pressure regulator, and fuel cell valves. Primary continuous-source target of 150 kW net at the DC link.
2. **High-Energy Battery (HEB):** 800V-class pack constructed from Amprius SA08 3.4V 11.05Ah Li-ion polymer pouch cells, managed via an n-BMS Battery Management System.
3. **Ultracapacitor Bank (SCB):** Maxwell BCAP3000 P300 K05 EDLC supercapacitors providing 500 kW burst capability for launch and regen absorption.
4. **Traction Motors & Drivetrain:**
    - **Rear Axle:** 2 × Helix SPX242-175D motors paired with CTI Silicon Carbide (SiC) inverters and Hewland drivetrain components.
    - **Front Axle:** 2 × Helix SPX242-94 motors paired with CTI Silicon Carbide (SiC) inverters.

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# 2. Complete Master Hardware Manifest

## Chassis & Structural

- **Chassis:** DASH TR01 Carbon-Fiber Monocoque (Dash-CAE) with ultra-high torsional rigidity (35–70 kNm/deg).

## Processing, VCU & Computing Hardware

- **Central HPC (ADAS & Infotainment):** Advantech MIC-743-AT7A1 (NVIDIA Jetson Thor AI Inference System) running SnappOS (Android Automotive OS) on a 17-inch High-Brightness Touchscreen.
- **Real-Time Zonal & Gateway Platforms:** NXP GoldBox 3 (Vehicle Networking) and NXP GreenBox 3 (Real-Time VDC Execution).
- **Safety VCU & Coprocessors:** AEM 30-8100 VCU 300 Programmable Electric Vehicle Control Unit and Teensy 4.1 boards equipped with SN65HVD230 CAN transceivers and MOSFET driver boards.

## High-Voltage (800V) Power Distribution & Safety

- **Contactors & Switching:** GIGAVAC GX110CEB Main Contactors, GIGAVAC P115CDA MiniTactors, and HIITIO Solid-State iBDUs.
- **Fusing & Protection:** Mersen MEV100A100-4 (1000V DC, 100A) EVpack-fuses, Eaton Bussmann high-speed square body fuses (170M6528), and low-voltage fuse links (EVKR60-700).
- **Insulation Monitoring & Surge Suppression:** Bender ISOMETER IR155-3204 Insulation Monitoring Devices and Bender IR155-3204 Surge Arrestors.
- **Charging & DC-DC Conversion:** NACS/Tesla DC Fast-Charge Inlet, Sensata EVCC, Ovar Tech Bidirectional 11kW OBC & 3kW DC-DC Converter (550-850VDC), Delta Electronics 800V On-Board Charger, and Sevcon Hi-V / Epic Power Converters EPC 18k 870i DC-DC Converters.
- **Busbars & Cabling:** Exxelia Powder-Coated High-Voltage Copper Busbars, TE Connectivity HVA 280 Automotive Housings, and HIVONEX YHV280 Automotive Housings.

## Low-Voltage (12V) Body & Zonal Architecture

- **Power Distribution:** Infinitybox Intelligent Power Management Kit (Mastercell and Powercell NGX).
- **Body Controllers:** Infinitybox inMOTION NGX, inCONTROL Rotary Interface, inLINK NGX Wireless, inCODE NGX System Programmer, inTOUCH NET Smart Device Interface, and inRESERVE Active BMS.

## Steering & Driver Interface

- **Steer-by-Wire & Assist:** DCE Dual MGU with autonomous control, DCE Contactless Torque and Angle Sensor.
- **Column & Switches:** Porsche Steering Column Assembly (Macan 95B419506T), Porsche Macan Column Switch (95B953502AH), 2020–2026 Porsche Multi-Function Switch (992-953-502-FK-HUO), and GT Sport Steering Wheel (992).

## Braking & Pedal Box

- **Pedal Assembly:** Tilton 600-Series Overhung-Mount Pedal Assemblies (72-615) with throttle linkage system and Penny & Giles TPS280DP Position Sensors.
- **Master Cylinders & Boosters:** Tilton 79-Series ABS-Compatible Master Cylinders, Tesla Brake Booster Assembly (1188671-00-A), Universal 4-Bolt Electric Power Brake Booster, and Adjustable Proportioning Blocks.
- **Calipers & Rotors:** AP Racing Mono E 10-piston front (CP7411) and 6-piston rear calipers, Tesla Parking Brake Caliper Kit (12V), Surface Transforms Carbon-Ceramic Brake Rotors, and Amphenol Brake Pressure Transducers.
- **ABS Integration:** Bosch Motorsport ABS M5 Kit ClubSport / Teves MK60 ABS Swap Kit.

## Thermal Management & HVAC

- **Cabin Climate:** Restomod Air Bantam-S Ultra Compact A/C Kit (with Variable Air Controllers, Quantum PODS controls, and Diablo Billet Vents), CEV Electric A/C Compressor Kit, and Vintage Air replacement components.
- **Heating:** Nanfeng Best NF OEM 7KW 800V PTC Coolant Heater.
- **Cooling & Pumping:** Davies-Craig EWP130 Electric Water Pumps, Davies-Craig Mustang EBP25 Boosters, 2020-2023 Tesla Y Water Pumps (Four Seasons 89079), Restomod Air Custom Condenser (16"x26"), and Zerex G40 Coolant.
- **Plumbing Hardware:** 5/8" ID Thick Brass Y-Shaped 3-Way Union Fittings and reinforced silicone/EPDM hosing.

## ADAS & Sensor Suite

- **Radar & LiDAR:** Smartmicro Extra Long Range Radar (DRVEGRD 171), Smartmicro UMRR-96 Radar, and InnovizTwo LiDAR Sensor.
- **Ultrasonic & Vision:** Valeo Mobility Kit Ultrasonic Sensors, Teledyne Blackfly S GigE Color Cameras, and Leopard Imaging LI-IMX390-GMSL2 Cameras.
- **Telemetry & IMU:** Bosch MM7.10 IMU, NovAtel GPS-702-GG-HV Pinwheel Antenna, MoTeC FIA Accident Data Recorder V2 (ADR2), Ecumaster CAN Thermal Cameras, Ecumaster CAN I/O Modules, Wheel Speed to CAN Modules, and Cerebrum ECO Smart Tire Sensors.

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# 3. Software Architecture & Network Stack

## Zonal Software-Defined Vehicle (SDV) Blueprint

- **Central HPC Layer:** NVIDIA Jetson Thor executing ADAS sensor fusion, vehicle dynamics control math (`helios-vdc`), energy management algorithms (`helios-energy`), and Android Automotive OS (SnappOS) via an eCAL high-throughput IPC backbone.
- **Zonal Communication Backbone:** Eclipse Cyclone DDS certified to ISO 26262 ASIL D, operating over Automotive Ethernet (1000BASE-T1 / 100BASE-T1) with IEEE 802.1AS (gPTP) time synchronization.
- **Zonal Node Firmware:** NXP S32K3 microcontrollers running `helios-vsm` (Vehicle State Machine) and `helios-zonal-fw` to bridge DDS commands to local actuators via CAN FD, PWM, and LIN.
- **Digital Twin Representation:** Eclipse Ibeji standardizing hardware state representations and telemetry models.

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# 4. Functional Safety Requirements & Fault-Response Matrix

## Functional Safety Goals (ASIL D Baseline)

- **SG1:** Prevent unintended motor acceleration exceeding $0.2g$ during cross-domain Ethernet/DDS communication jitter or loss.
- **SG2:** Ensure immediate high-voltage isolation within $2\text{ ms}$ upon breach of the High-Voltage Interlock Loop (HVIL) or detection of an active insulation fault via the Bender ISOMETER.
- **SG3:** Prevent asymmetric braking forces exceeding yaw-rate error tolerances ($> 5^\circ/\text{s}$ at speed) during regenerative and friction brake blending failures.

## Fault-Response Table

| Fault ID | Fault Description | Diagnostic Detection Time | Safety Reaction | Fault Action / State Transition |
|---|---|---|---|---|
| **F_HPC_HEARTBEAT** | Loss of Jetson Thor / DDS heartbeat on Automotive Ethernet | $\le 10\text{ ms}$ | Transition NXP GreenBox / AEM VCU to local safety mode | Zero torque command to CTI inverters; gently engage regenerative braking via rear motors. |

| Fault ID | Fault Description | Diagnostic Detection Time | Safety Reaction | Fault Action / State Transition |
|---|---|---|---|---|
| **F_HVIL_BREACH** | High-Voltage Interlock Loop continuity broken at connectors (TE HVA 280 / HIVONEX) | $\le 2\text{ ms}$ | Hardware-level trip of GIGAVAC GX110CEB main contactors | Open main battery disconnects; bleed intermediate circuit via pre-charge/discharge path. |
| **F_IMD_ISOLATION** | Insulation resistance drops below $500,\Omega/\text{V}$ (Bender IR155) | $\le 100\text{ ms}$ | Controlled high-voltage system shutdown and isolation | Ramp down fuel cell and battery output; log insulation fault to non-volatile memory; prevent re-arm. |
| **F_INVERTER_MISMATCH** | CTI Silicon Carbide inverter torque feedback mismatch ($\Delta > 10%$) | $\le 5\text{ ms}$ | Immediate inverter hardware disable on faulted axle | Open local axle contactors; maintain stable steering via DCE Dual MGU. |
| **F_BRAKE_PRESSURE** | Hydraulic pressure divergence or ABS failure (Bosch Motorsport M5 / Tilton Master Cylinders) | $\le 5\text{ ms}$ | Fallback to mechanical/electronic parking brake caliper backup | Engage secondary braking circuits; illuminate dashboard warning flag on display. |

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